Human Computer Interaction

First Part

Human computer interaction

1- Introduction
2- Why HCI
3- What is HCI
4- Define the HCI
5- Generations interaction process

Introduction

- **Designing interactive computer systems to be effective, efficient, easy, and enjoyable *to use* is important, so that people and society may realize the benefits of computation based devices.**

- **human computer interaction is a specialty in many fields, and is therefore multidisciplinary, but it has an intrinsic relationship as a subfield to computer science.**





- **Most interactive computing systems are for some human purpose and interact with humans in human contexts.**
- **In fact, that largest part of program code in most interactive systems deals with user interaction.**
- **Inadequate attention to users and task context not only leads to bad user interfaces, it puts entire systems at risk.**



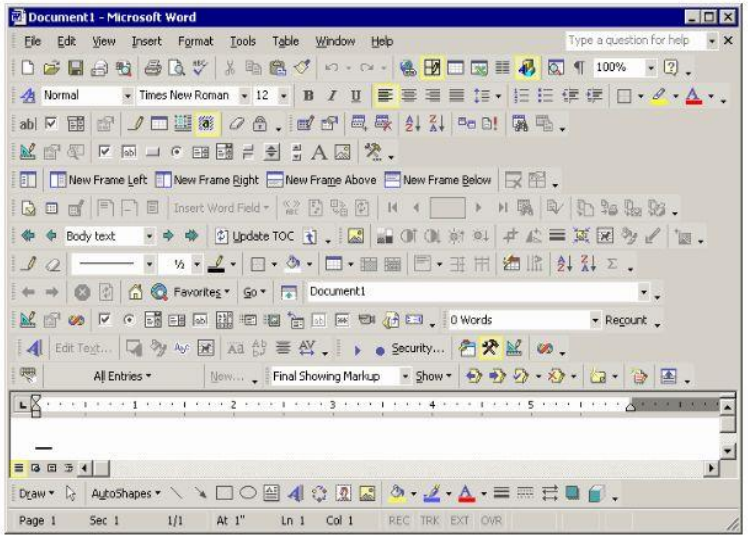


Why HCI

- Computer and related devices have to be designed with an understanding that people with specific tasks in mind will want to use them in a way that is seamless with respect to their everyday work.

- To do this, those who design these systems need to know how to think in terms of the eventual users' task and how to translate that knowledge into an executable system.
- The interface is not something that can be plugged in at the last minute; its design should be developed integrally with the rest of the system.

It should not just present a 'pretty face',
but should support the tasks that
people actually want to do,
and forgive the careless mistakes.



What is HCI

- The term human computer interaction has only been in widespread use since the early 1980s, but has its roots in more established disciplines.

- The Second World War provided the impetus for studying the interaction between human and machines, as each side strove to produce more effective weapons systems. This led to a wave of interest in area among researchers, and formation of the Ergonomics Research Society in 1949.



- This research originally went under the name man machine interaction, but this become human computer interaction in recognition of the particular interest in computers and the composition of the user population.

- **Another stand of research that has influenced the development of HCI is information science and technology.**

- **However, when we about human computer interaction, we do not necessarily envisage a single user with a desktop computer. By user we many mean an individual user, a group of users working together, or a sequence of users in an organization, each dealing with some part of the task or process. The user is whoever is trying to get the job done using the technology.**



By interaction we mean any communication between a user and computer, be it direct or indirect.

- **Direct interaction involves a dialog with feedback and control throughout performance of the task.**

- **Indirect interaction may involve batch processing or intelligent sensors controlling the environment .**

The important thing is that the user is interacting with the computer in order to accomplish something.



Direct manipulation

There are three "use" words that must all be true for a product to be successful ; it must be:

1- Useful : accomplish what is required.

2- Useable: do it easily and naturally, without danger of error.

3- Used : make people want to use it, be attractive, engaging, fun.

Define the HCI

Human Computer Interaction

HCI the study of how humans interact with computers, and how to design computer systems that are easy, quick and productive for humans to use.

Human-Computer Interaction

HCI is the study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.

Human Computer Interaction

Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them

Define the Interactive Interface

Interactive Interface

As a mediator, which carries user actions through the implementation of the work of the technical help that the implementation of this task.

Interactive Interface

It is a collection "informatics" tools that allow a user or group of users to do something by using a technique intended to carry out this work.



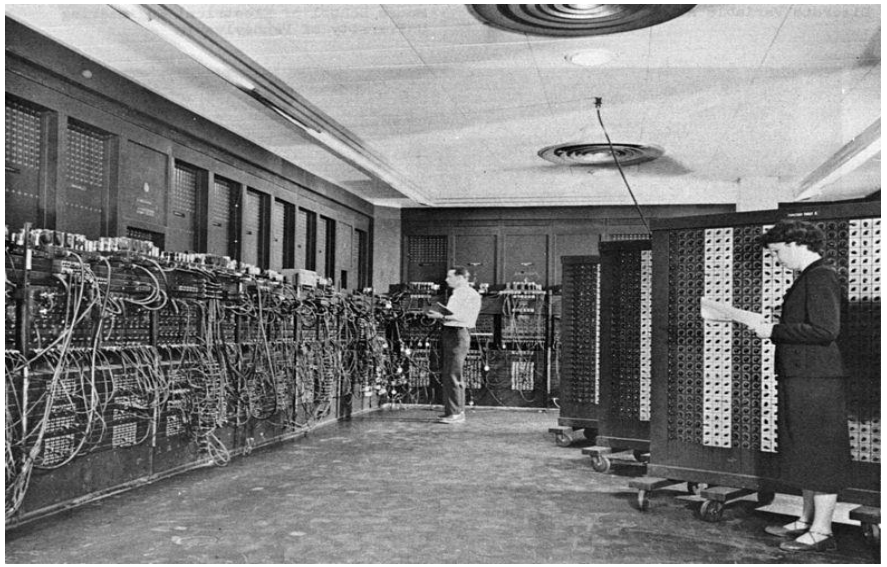
Stages of the development interaction between human and computer

Stages of the development process of the interaction between human and computer

First generation (1945 - 1956):

1- There was no interaction between the human-computer

2- Computers were limited to only experts and specialists.



Electronic Numerical Integrator and Computer ENIAC 1946

Second and third generation (1956 - 1970):

1- **Separation between software and solid construction**

2- The emergence of operating systems FMT, VMS, DOS and programming languages at a high level. (COBOL, FORTRAN).

3-The interaction between man and computer is very limited.

4- The emergence of input and output tools reader and perforated cards, and printers.

5- Computers were very expensive and is still confined to specialists



Grace Hopper

the inventor of compiler

- **Compiler improved usability**
- **COBOL 1959**



Sketchpad 1963

it could draw both horizontal and vertical lines and combine them into figures and shapes. figures could be copied, moved, rotated, or resized, retaining their basic properties

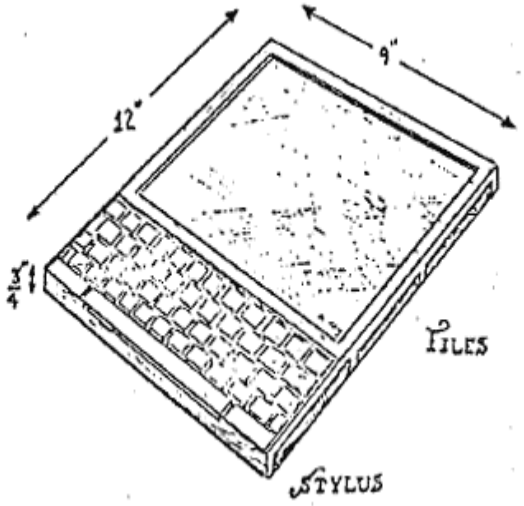


computer mouse 1964

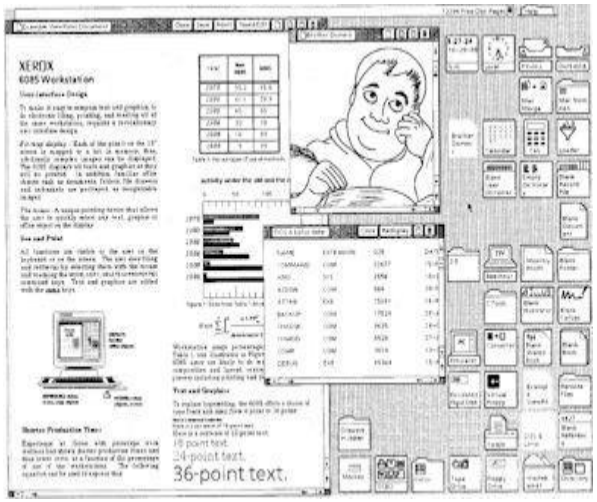
demonstrated the interaction using the mouse at the mother of all demos

Modern computers (1971- …..)

Informatics applications have evolved and emerged new forms of interaction between humans and computers and dropped price of computers has become so ordinary people can possess.



Dynabook : 1972
concept of laptop



Graphical user interface
XEROX Star 1981
Microcomputers



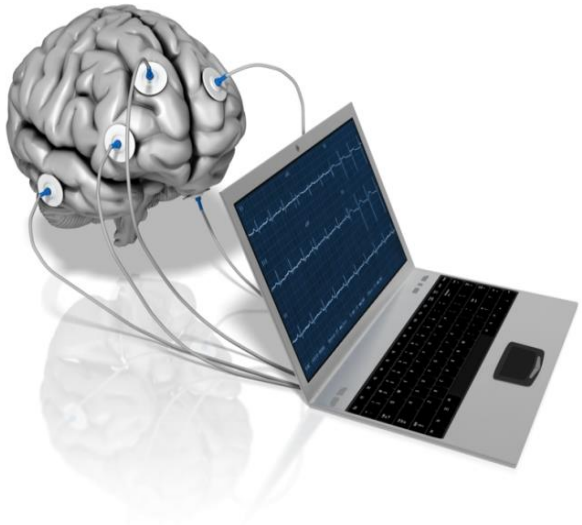
Motorola released 1983
Motorola DynaTAC 8000X

Human Computer Interaction

Second Part

Human Factor

1- Human's Input output Channels
2- Human memory
3- Thinking : Reasoning and Problem Solving

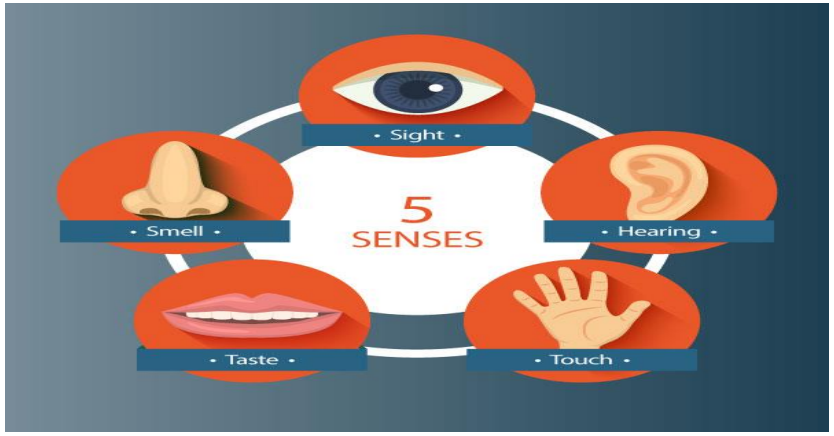


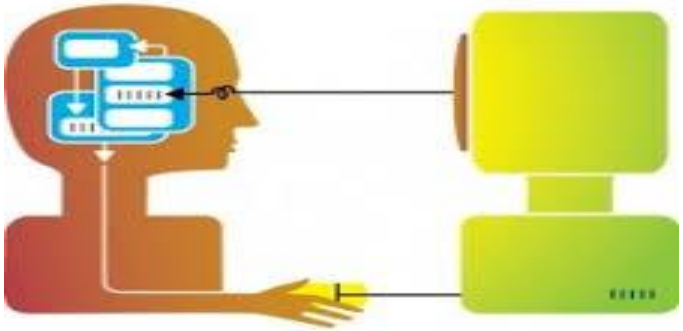
Human's Input output Channels

- A person's interaction with the outside world occurs through information begin received and sent; input and output.

- In an interaction with a computer the user receives information that is output by the computer, and responds by providing input to the computer the user's output becomes the computer's input and vice versa.

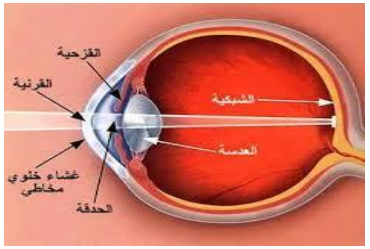
- There are five major senses; sight, hearing, touch, taste and smell. Of these, the first three are the most important to HCI. Taste and smell do not currently play a significant role in HCI.





Vision

Human vision is a highly complex activity with a range of physical and perceptual limitations, yet it is the primary source of information for the average person.



We can roughly divide visual perception into two stages:

1- The physical reception of the stimulus from the outside world

2- The processing and interpretation of that stimulus.

Vision begins with light. The eye is a mechanism for receiving light and transforming it into electrical energy.

There are several stages in the reading process.

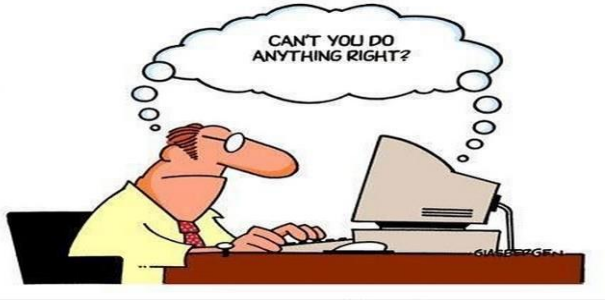
1- **First the visual pattern of the word on the page iv perceived .**

2- Second it is then decoded with reference to an internal representation of language.

3- Third language processing include syntactic and semantic analysis and operate on phrases or sentences.

The eye moves backwards over the text as well as forwards, in what are known as regressions.

If the text is complex there will be more regressions.



Hearing

- The sense of hearing is often considered secondary to sight, but we tend to underestimate the amount of information that we receive through our ears.

- sound can convey a remarkable amount of information. It is rarely used to its potential in interface design, usually being confined to warning sounds and notifications. The exception is multimedia, which may include music, voice commentary and sound effects. This suggests that sound could be used more extensively in interface design, to convey information about the system state.



Touch

The third and last of the senses that we will consider is touch or haptic perception. Although this sense is often viewed as less important than sight or hearing, imagine life without it.

- **Touch provides us with vital information about our environment. It tells us when we touch something hot or cold, and can therefore act as a warning**
- **The apparatus of touch differs from that of sight and hearing in that it is not localized.**
- **We receive stimuli through the skin. The skin contains three types of sensory receptor.**
- **although the whole of the body contains such receptors, some areas have greater sensitivity or acuity than others.**



Movement

The stimulus (of the question) is received through the sensory receptors and transmitted to the brain. the question is processed and a valid response generated. the brain the tells the appropriate muscles to respond. each of these stage take time, which can be roughly divided into reaction time and movement time

- **Movement time: is dependent largely on the physical characteristics of the subject ; their age and fitness.**
- **Reaction time varies according to the sensory channel through which the stimulus is received.**
- **factors such as skill or practice can reduce reaction time, and fatigue can increase it.**
- **Speed and accuracy of movement are important considerations in the design of interactive systems, primarily in terms of the time taken to move to a particular target on a screen.**
- **In some cases requiring increased reaction time reduces accuracy.**

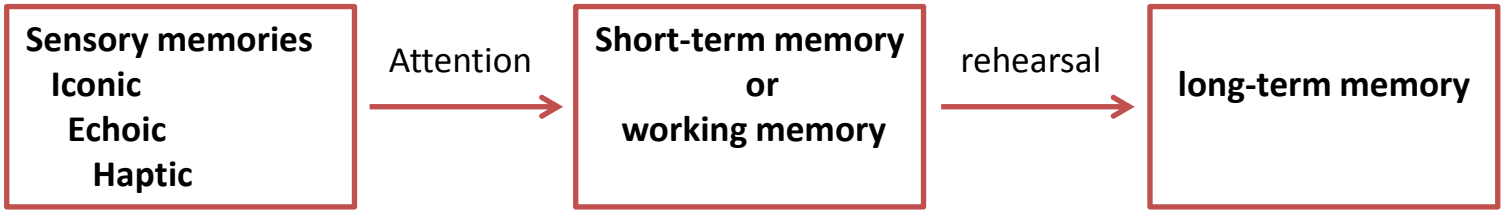


Human Memory

Indeed, much of our everyday activity relies on memory. as well as storing all our factual knowledge, our memory contains our knowledge of actions or procedures. It allows us to repeat actions, to use language, and to use new information received via our senses. It also gives us our sense of identity, by preserving information from our past experiences.

❖ **It is generally agreed that there are three types of memory or memory function:**

1. **Sensory buffers**
2. **Short time memory (working memory)**
3. **Long term memory**



A model of the structure of memory

1- Sensory memory

The sensory memories act as buffers for stimuli received through the senses. A sensory memory exists for each sensory channel:

1- Iconic memory for visual stimuli.

2- Echoic memory for aural stimuli.

3- Haptic memory for touch.

These memories are constantly overwritten by new information coming in on these channels.

2- Short -term memory

Acts as a " scratch-pad" for temporary recall of information. It is used to store information which is only required fleetingly.

- **short term memory can be accessed rapidly, in the order of 70 ms. however, it also decays rapidly, meaning that information can only be held there temporarily**
- **Short term memory has a limited capacity.**
- **There are two basic methods for measuring memory capacity.**

1- Length of a sequence which can be remembered in order.

2- The second allows items to be freely recalled in any order.

3- Long -term memory

Is our main resource. Here we store factual information, experiential knowledge, procedural rules of behavior in fact, everything that we 'know'.
It differs from short-term memory in a number of significant ways:

1. **It has a huge if not unlimited, capacity**
2. **It has a relatively slow access time of approximately a tenth of a second.**
3. **Forgetting occurs more slowly in long-term memory.**

Long-term memory is intended for the long-term stage of information. Information is placed there from working memory through rehearsal. Unlike working memory there is little decay: long-term recall after minutes is the same as that after hours or days.

there are two types of long-term memory :

1. **Episodic memory**
2. **Semantic memory**

Thinking : Reasoning and Problem Solving

This is perhaps the area which is most complex and which separates humans from other information processing system, both artificial and natural.

- **Although it is clear that animals receive and store information, there is little evidence to suggest that they can use it in quite the same way as humans.**
- **Similarly, artificial intelligence has produced machines which can see and store information. But their ability to use that information is limited to small domains.**

Thinking can require different amounts of knowledge. Some thinking activities are very directed and the knowledge required is constrained. others require vast amount of knowledge from different domains.

Reasoning

is the process by which we use the knowledge we have to draw conclusions or infer something new about the domain of interest. There are a number of different types of reasoning. We use each of these types of reasoning in everyday life, but they differ in significant way.

1- Deductive reasoning
2- Inductive reasoning
3- Abductive reasoning

Problem solving

If reasoning is a means of inferring new information from what is already known

- **Problem solving is the process of finding a solution to an unfamiliar task, using the knowledge we have.**

Human problem solving is characterized by the ability to adapt the information we have to deal with new situation.
There are a number of different views of how people solve problems:

1. **Gestalt theory**
2. **Problem space theory**
3. **Analogy in problem solving**

Human Computer Interaction

Third Part

Computer Factor

1- Typical Computer System
2- Input / output Devices
3- Memory
4- Limitations on interactive performance



Computer System

- In the early days of computing, information was entered into the computer in a large mass – batch data entry. There was minimal interaction with the machine: the user would simply dump a pile of punched cards onto a reader, press the start button, and then return a few hours later.
- This still continues today although now with pre-prepared electronic files or possibly machine-read forms. It is clearly the most appropriate mode for certain kinds of application, for example printing pay checks or entering the results from a questionnaire.



The Entry Device

- Entering text is one of our main activities when using the computer. The most obvious means of text entry is the plain keyboard, but there are several variations on this:

Alphanumeric keyboard : Is still one of the most common input devices in use today. It is used for entering textual data and commands.

Chord keyboards: are significantly different from normal alphanumeric keyboards. Only a few keys, four or five, are used and letters are produced by pressing one or more of the keys at once.

Phone pad and T9 entry with mobile phones being used for SMS text messaging the phone keypad has become an important form of text input.







G7 Chord in Casio Keyboards

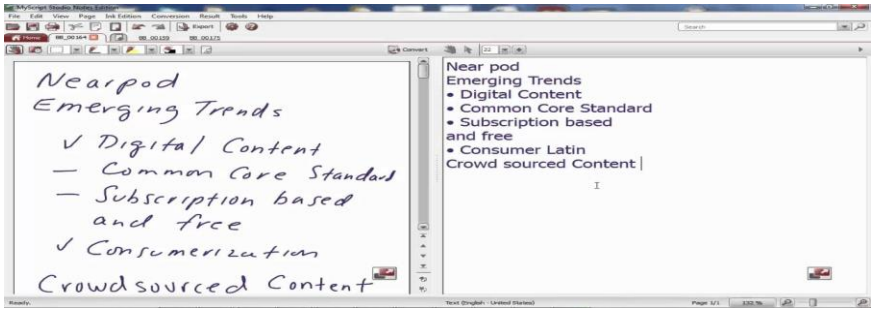


Handwriting Recognition

Handwriting is a common and familiar activity, and is therefore attractive as a method of text entry. If we were able to write as we would when we use paper, but with the computer taking this form of input and converting it to text, we can see that it is an intuitive and simple way of interacting with the computer.

Speech recognition

is a promising area of text entry, but it has been promising for a number of years and is still only used in very limited situations. There is a natural enthusiasm for being able to talk to the machine and have it respond to commands, since this form of interaction is one with which we are very familiar.





POSITIONING, POINTING AND DRAWING

Pointing devices allow the user to point, position and select items, either directly or by manipulating a pointer on the screen. Many pointing devices can also be used for free-hand drawing although the skill of drawing with a mouse is very different from using a pencil. The mouse is still most common for desktop computers.





1- The mouse
2- Touchpad
3- Trackball and thumbwheel
4- Joystick and keyboard nipple
5- Touch-sensitive screens
6- Stylus and light pen
7- Digitizing tablet











Output Device

1- Display Devices

The vast majority of interactive computer systems would be unthinkable without some sort of display screen.

Virtually all computer displays are based on some sort of bitmap. That is the display is made of vast numbers of colored dots or pixels in a rectangular grid. These pixels may be limited to black and white in grayscale, or full color.

2- Physical Controls, Sensors and Special Devices

These devices may have special displays, may use sound, touch and smell as well as visual displays, may have dedicated controls and may sense the environment or your own bio-signs.

1- Special displays

2- Sound output

3- Environment and bio-sensing

3- Printing Devices

They print processed data in a form humans can read. They are the most commonly used output devices. To produce the output, the printers first receive electronic signals from CPU. Printers vary greatly in performance and design. Printers are divided into two main set.

1- Impact Printers

That use electromechanical mechanisms that cause hammers to strike against a ribbon and the paper. (dot matrix)

2- Non Impact Printers

That do not hit or impact a ribbon to print, but use heat, laser technology or photographic techniques to print output. (inkjet, laser)

Scanners and optical character recognition

Printers take electronic documents and put them on paper – scanners reverse this process. They start by turning the image into a bitmap, but with the aid of optical character recognition can convert the page right back into text. The image to be converted may be printed, but may also be a photograph or hand-drawn picture. There are two main kinds of scanner: flat-bed and hand-held.

1- Fat-bed scanner
2- Hand-held scanners





Memory

- Like human memory, we can think of the computer's memory as operating at different levels, with those that have the faster access typically having less capacity.
- By analogy with the human memory, we can group these into short-term and long-term memories (STM and LTM), but the analogy is rather weak – the capacity of the computer's STM is a lot more than seven items!
- The different levels of computer memory are more commonly called primary and secondary storage.

1- RAM and short-term memory (STM)

- At the lowest level of computer memory are the registers on the computer chip, but these have little impact on the user except in so far as they affect the general speed of the computer.
- Most currently active information is held in silicon-chip random access memory (RAM).
- Different forms of RAM differ as to their precise access times, power consumption and characteristics.

2- Disks and long-term memory (LTM)

- For most computer users the LTM consists of disks, possibly with small tapes for backup. The existence of backups, and appropriate software to generate and retrieve them, is an important area for user security.
- There are two main kinds of technology used in disks: magnetic disks and optical disks.
- The most common storage media, floppy disks and hard (or fixed) disks, are coated with magnetic material, like that found on an audio tape, on which the information is stored.
- With disks there are two access times to consider, the time taken to find the right track on the disk, and the time to read the track.

Human Computer Interaction

Fifth Part

What is Interaction Design?

1- What is design
2- The golden rule of design
3- To err is human
4- The process of design
5- Interaction Style
6- Elements of the WIMP Interface

What is design

So what is design? A simple definition is:

achieving goals within constraints

This does not capture everything about design, but helps to focus us on certain things:

Goals :

What is the purpose of the design we are intending to produce? Who is it for? Why do they want it?

Constraints:

- **What materials must we use?**
- **What standards must we adopt?**
- **How much can it cost?**
- **How much time do we have to develop it?**
- **Are there health and safety issues?**

Of course, we cannot always achieve all our goals within the constraints. So perhaps one of the most important things about design is:

Trade-off

Choosing which goals or constraints can be relaxed so that others can be met. Often the most exciting moments in design are when you get a radically different idea that allows you to satisfy several apparently contradictory constraints. However, the more common skill needed in design is to accept the conflict and choose the most appropriate trade-off. The temptation is to focus on one or other goal and optimize for this, then tweak the design to make it just satisfy the constraints and other goals. Instead, the best designs are where the designer understands the trade-offs and the factors affecting them. Paradoxically, if you focus on the trade-off itself the more radical solutions also become more apparent.

The golden rule of design

The designs we produce may be different, but often the raw materials are the same. This leads us to the golden rule of design:

understand your materials

For Human–Computer Interaction the obvious materials are the human and the computer. That is we must:

- **understand computers**
  limitations, capacities, tools, platforms

- **understand people**

psychological, social aspects, human error.

To err is human

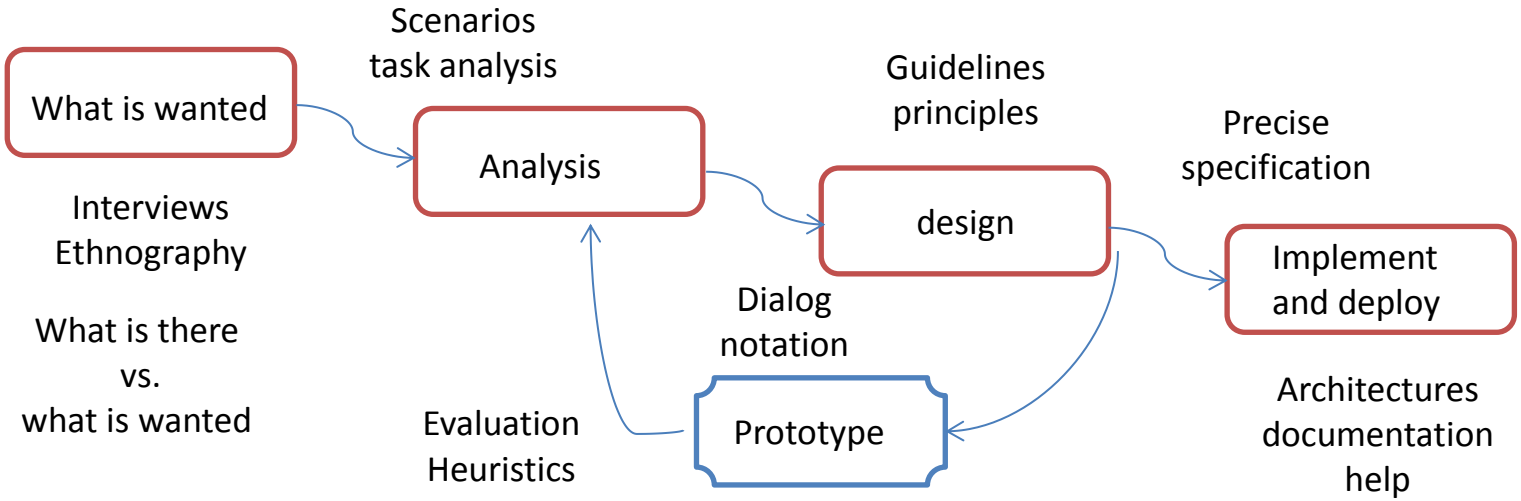
It might sound demeaning to regard people as 'materials', possibly even dehumanizing. In fact, the opposite is the case: physical materials are treated better in most designs than people. This is particularly obvious when it comes to failures.
The phrase 'human error' is taken to mean 'operator error', but more often than not the disaster is inherent in the design or installation of the human interface. Bad interfaces are slow or error-prone to use. Bad interfaces cost money and cost lives.

Similarly it is the nature of humans to make mistakes, and systems should be designed to reduce the likelihood of those mistakes and to minimize the consequences when mistakes happen.

- **Often when an aspect of an interface is obscure and unclear, the response is to add another line in the manual.**
- **If you design using a physical material, you need to understand how and where failures would occur and strengthen the construction, build in safety features or redundancy.**
- **If you treat the human with as much consideration as a piece of steel or concrete, it is obvious that you need to understand the way human failures occur and build the rest of the interface accordingly.**

The process of design

Often HCI professionals complain that they are called in too late. A system has been designed and built, and only when it proves unusable do they think to ask how to do it right! In other companies usability is seen as equivalent to testing checking whether people can use it and fixing problems, rather than making sure they can from the beginning. Here we'll take a simplified view of four main phases plus an iteration loop, focused on the design of interaction



Interaction design process

1- What is wanted



The first stage is establishing what exactly is needed. As a precursor to this it is usually necessary to find out what is currently happening. What sort of personal appliances do they currently use? There are a number of techniques used for this in HCI: interviewing people, videotaping them, looking at the documents and objects that they work with, observing them directly.

2- Analysis



The results of observation and interview need to be ordered in some way to bring out key issues and communicate with later stages of design. which are a means to capture how people carry out the various tasks that are part of their work and life. we will look at scenarios, rich stories of interaction, which can be used in conjunction with a method like task analysis or on their own to record and make vivid actual interaction. These techniques can be used both to represent the situation as it is and also the desired situation

3- Design



There are numerous rules, guidelines and design principles that can be used to help with this. We need to record our design choices in some way and there are various notations and methods to do this, including those used to record the existing situation. It is at this stage also where input from theoretical work is most helpful, including cognitive models, organizational issues and understanding communication.

4- Iteration and prototyping



Humans are complex and we cannot expect to get designs right first time. We therefore need to evaluate a design to see how well it is working and where there can be improvements. Some forms of evaluation can be done using the design on paper, but it is hard to get real feedback without trying it out. Most user interface design therefore involves some form of prototyping, producing early versions of systems to try out with real users.

5- Implementation and deployment

Finally, when we are happy with our design, we need to create it and deploy it. This will involve writing code, perhaps making hardware, writing documentation and manuals – everything that goes into a real system that can be given to others.

- **there is a trade-off between the length of the design period and the quality of the final design. This means one sometimes has to accept a design as final even if it is not perfect: it is often better to have a product that is acceptable but on time and to cost than it is to have one that has perfect interaction but is late and over budget.**

Interaction Style

Interaction can be seen as a dialog between the computer and the user. The choice of interface style can have a profound effect on the nature of this dialog. Here we introduce the most common interface styles and note the different effects these have on the interaction. There are a number of common interface styles including

1- Command line interface
2- Menus
3- Natural language
4- Question/answer and query dialog
5- Form-fills and spreadsheets
6- WIMP
7- Point and click
8- Three-dimensional interfaces.

1- Command line interface

The command line interface was the first interactive dialog style to be commonly used and, in spite of the availability of menu-driven interfaces, it is still widely used. It provides a means of expressing instructions to the computer directly, using function keys, single characters, abbreviations or whole-word commands

```
sable.soc.staffs.ac.uk> javac HelloWorldApp
javac: invalid argument: HelloWorldApp
use: javac [-g][-O][-classpath path][-d dir] file.java…
sable.soc.staffs.ac.uk> javac HelloWorldApp.java
sable.soc.staffs.ac.uk> java HelloWorldApp
Hello world!!
sable.soc.staffs.ac.uk>
```

Figure 3.7 Command line interface

2- Menus

In a menu-driven interface, the set of options available to the user is displayed on the screen, and selected using the mouse, or numeric or alphabetic keys. Since the options are visible they are less demanding of the user, relying on recognition rather than recall.

```
PAYMENT DETAILS                P3-7

please select payment method:
  1. cash
  2. check
  3. credit card
  4. invoice

  9. abort transaction
```

Figure 3.8 Menu-driven interface

3- Natural language

Perhaps the most attractive means of communicating with computers, at least at first glance, is by natural language. Users, unable to remember a command or lost in a hierarchy of menus, may long for the computer that is able to understand instructions expressed in everyday words! Natural language understanding, both of speech and written input, is the subject of much interest and research

4- Question/answer and query dialog

Question and answer dialog is a simple mechanism for providing input to an application in a specific domain. The user is asked a series of questions (mainly with yes/no responses, multiple choice, or codes) and so is led through the interaction step by step. An example of this would be web questionnaires.



Figure 3.9 A typical form-filling interface. Screen shot frame reprinted by permission from Microsoft Corporation

5- Form-fills and spreadsheets

Form-filling interfaces are used primarily for data entry but can also be useful in data retrieval applications. The user is presented with a display resembling a paper form, with slots to fill in. Often the form display is based upon an actual form with which the user is familiar, which makes the interface easier to use



Figure 3.11 A typical UNIX windowing system – the OpenLook system. Source: Sun Microsystems, Inc.

6- WIMP

Currently many common environments for interactive computing are examples of the WIMP interface style, often simply called windowing systems. WIMP stands for windows, icons, menus and pointers (sometimes windows, icons, mice and pull-down menus), and is the default interface style for the majority of interactive computer systems in use today, especially in the PC and desktop workstation arena.

7- Point and click

In most multimedia systems and in web browsers, virtually all actions take only a single click of the mouse button. You may point at a city on a map and when you click a window opens, showing you tourist information about the city. You may point at a word in some text and when you click you see a definition of the word.



8- Three-dimensional interfaces.

There is an increasing use of three-dimensional effects in user interfaces. The most obvious example is virtual reality, but VR is only part of a range of 3D techniques available to the interface designer.



Elements of the WIMP Interface

We have already noted the four key features of the WIMP interface that give it its name (windows, icons, menus , and pointers) and we will now describe these in turn. There are also many additional interaction objects and techniques commonly used in WIMP interfaces, some designed for specific purposes and others more general. We will look at buttons, toolbars, palettes and dialog boxes.

1- Windows
2- Icons
3- Menus
4- Pointers
5- Buttons
6- Toolbars
7- Palettes
8- Dialog boxes .

Together, these elements of the WIMP interfaces are called widgets, and they comprise the toolkit for interaction between user and system